Formulaite UI/UX Report

A survey of the Formulaite platform's design language, color system, and an in-depth walkthrough of its AI-driven formulation ("blending") tool. Prepared from a live hands-on session at app.formulaite.ai.

1. Overview & Methodology

This report is based on a live walkthrough of the Formulaite platform, a web application aimed at cosmetic/supplement R&D teams. The session covered the landing/tool-selection screen, a full end-to-end run of the "R&D Agent" chat-based formulation builder (building a vitamin C brightening serum from scratch), the Saved Formulations workspace, and the entry point to the separate "Simulaite Engine" simulation suite. Screenshots were captured at each meaningful stage and are embedded throughout this document.

2. Overall Visual Design & Color Palette

Formulaite uses a dark, lab-themed interface that reads as clean and scientific rather than playful. Typography is set in Inter (with a standard system-ui fallback stack), giving the product a modern, neutral, SaaS-dashboard feel. Spacing is generous, borders are subtle (thin, low-opacity white strokes rather than heavy drop shadows), and a single accent color is used consistently for anything interactive, which keeps the eye anchored despite a fairly dense amount of on-screen information. Inspecting the live CSS during the session surfaced the following concrete palette:

- Page background: rgb(11, 13, 19) / ~#0B0D13 — a near-black navy, not pure black, which softens contrast slightly
- Card / secondary surface: rgb(26, 29, 40) / ~#1A1D28 — a slightly lighter charcoal used for header pills and secondary buttons
- Primary accent (green): rgb(16, 185, 129) / #10B981 — Tailwind's "emerald-500"; used for the logo leaf, active states, the Plans badge, links, and compliance checkmarks
- Deep green fill: rgb(13, 77, 63) / ~#0D4D3F — used for filled primary buttons (e.g. "Start Chat") to contrast with the lighter emerald used for text/icons
- Heading text: rgb(228, 229, 233) / ~#E4E5E9 — an off-white rather than pure white, easier on the eyes on the dark background
- Body / muted text: rgb(107, 114, 128) and rgb(156, 163, 175) / ~#6B7280 and #9CA3AF — two steps of gray used for descriptions, timestamps, and placeholder copy
- Secondary indigo accent: rgba(99, 102, 241, 0.25) — appears occasionally (e.g. certain avatar/icon fills), breaking the green monochrome very sparingly

The result is a restrained, almost monochrome dark theme where green alone is left to signal interactivity and status (success, compliance, progress). This works well for scannability, though it also means the manufacturer-partner promotional banners (see Section 6) are styled in the exact same reassuring green as genuine system feedback (e.g.

the “Compliant” badge), which makes it easy to mistake a marketing unit for a scientific result at a glance.

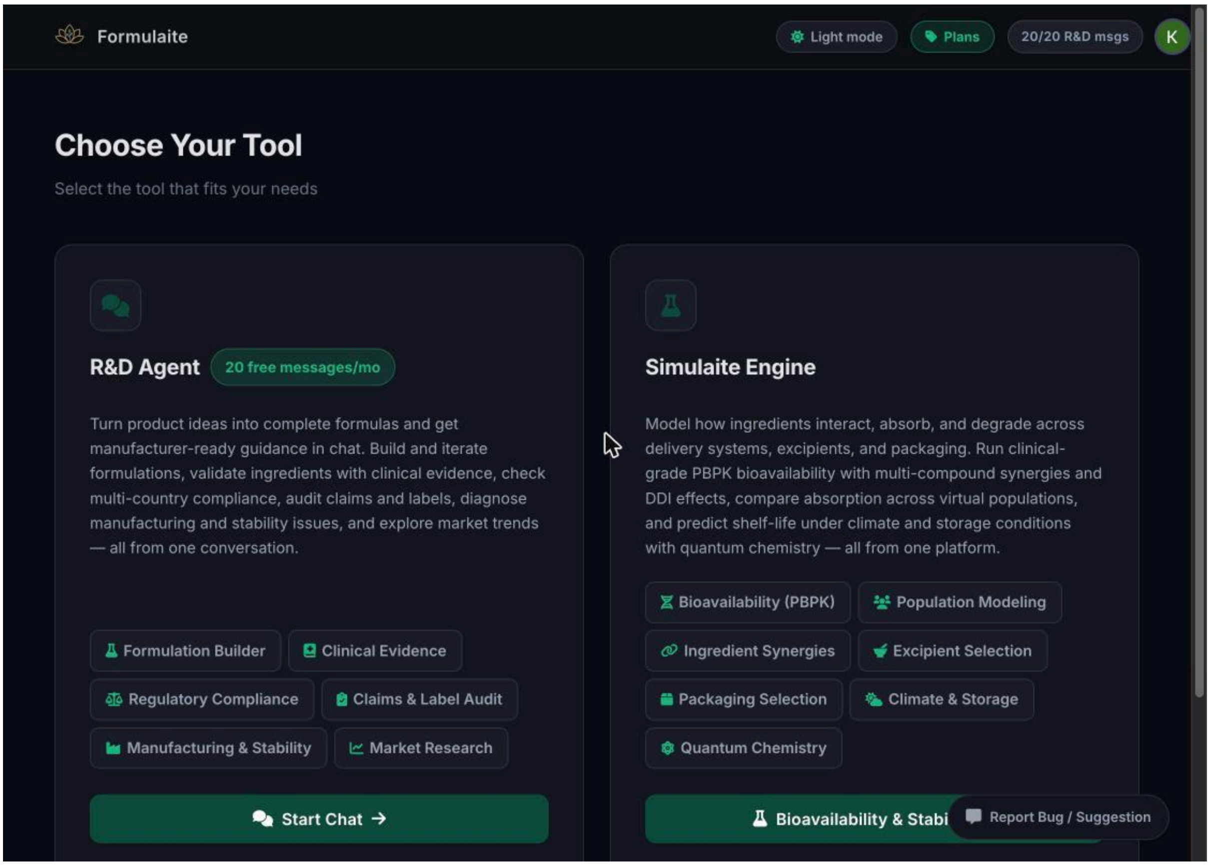

Figure 1. The landing “Choose Your Tool” screen, showing the two top-level tools (R&D Agent chat vs. Simulaite Engine) and the dark, green-accented visual language.

3. Navigation & Structure

The app opens on the “Choose Your Tool” screen (Figure 1) rather than a dashboard, presenting two large side-by-side cards: R&D Agent (a conversational formulation assistant) and Simulaite Engine (an advanced simulation suite covering bioavailability/PBPK, population modeling, ingredient synergies, excipient and packaging selection, climate/storage, and quantum chemistry). Each card lists its sub-capabilities as small pill-shaped tags that double as shortcuts into that specific sub-mode — a sensible pattern for a multi-tool platform, since it sets expectations before the user commits to a workflow.
One notable gap: the Simulaite Engine card’s primary call-to-action does not open a tool at all. Clicking it (or the “Plans” link in the header) routes straight to a pricing/enterprise contact page, so a free-tier user hits a paywall immediately rather than seeing even a preview of that engine’s own interface. That distinction — chat tool freely accessible, simulation engine enterprise-gated — is not communicated on the card itself.

4. The Formulation / Blending Tool — Detailed Walkthrough

This is the core of the platform and the focus of this report. It is best described as a guided, conversational formulation wizard rather than a manual ingredient-blending interface with sliders or percentage fields. The full flow, walked through live, is broken down below.

4.1 Entry point & guided intake

The chat opens with a welcome message listing everything the agent can do (build formulas, validate ingredients, audit compliance/claims, diagnose manufacturing issues, etc.), plus quick-start suggestion chips (e.g. “Build me a sleep capsule formula”) so a new user never faces an intimidating blank box. Describing a product in free text (I typed “Build me a vitamin C brightening face serum formula”) does not immediately produce an answer.

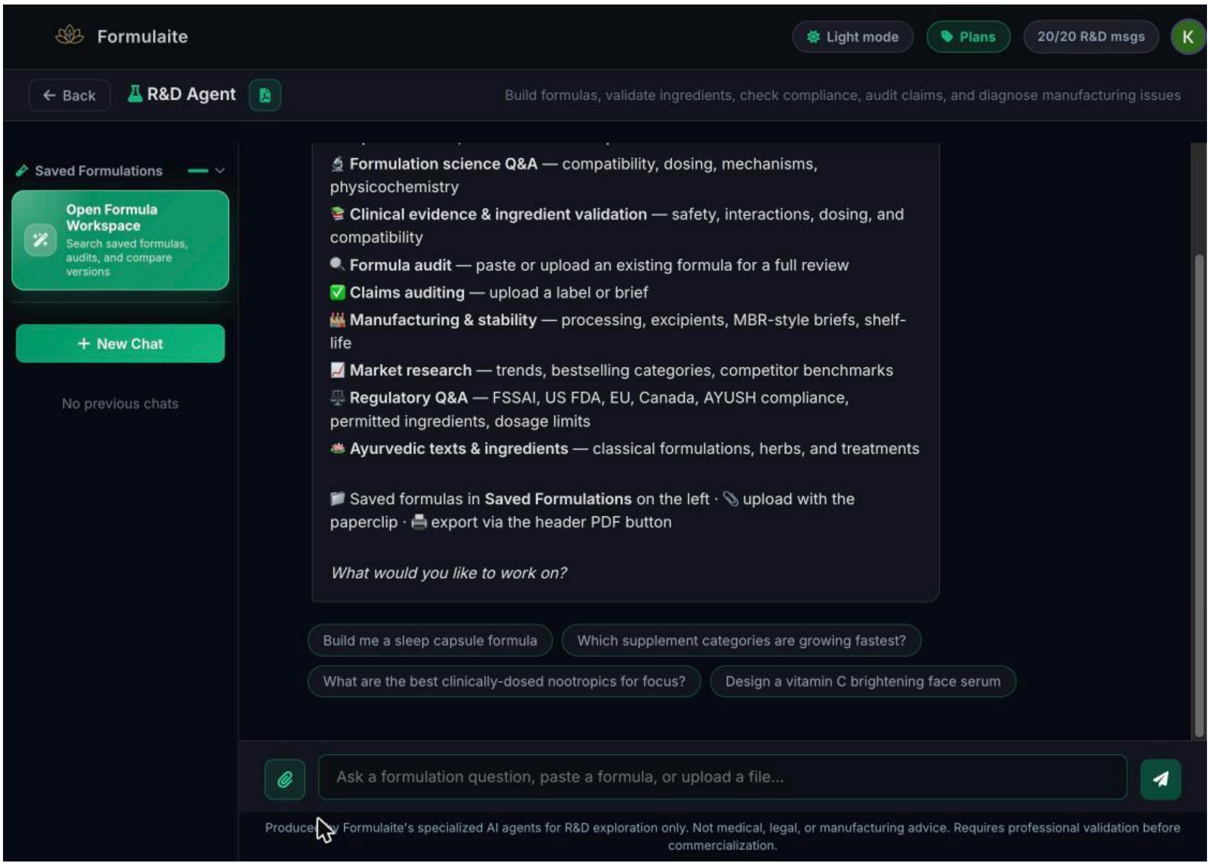

Figure 2. R&D Agent chat opening state: capability list plus quick-start suggestion chips, and the Saved Formulations panel on the left.

4.2 Adaptive clarifying questions before generation

Rather than generating a formula on the first message, the agent asks a short sequence of clarifying questions one at a time: number of active ingredients, target demographic, ingredient constraints (natural/herbal/none), cost tier (premium/standard/budget), and which regulatory frameworks to check. Each question pairs explanatory bullet text with clickable quick-reply chips, so answers rarely require typing. One question used a distinct widget type entirely: multi-select country/flag toggles for compliance frameworks (India, United States, Canada, EU) with an explicit “Confirm” button and a “No compliance checks” skip option below it. This shows the interface adapting its input control to the data being collected rather than reusing one generic text box throughout — a good micro-interaction choice for a form that is otherwise all free chat.

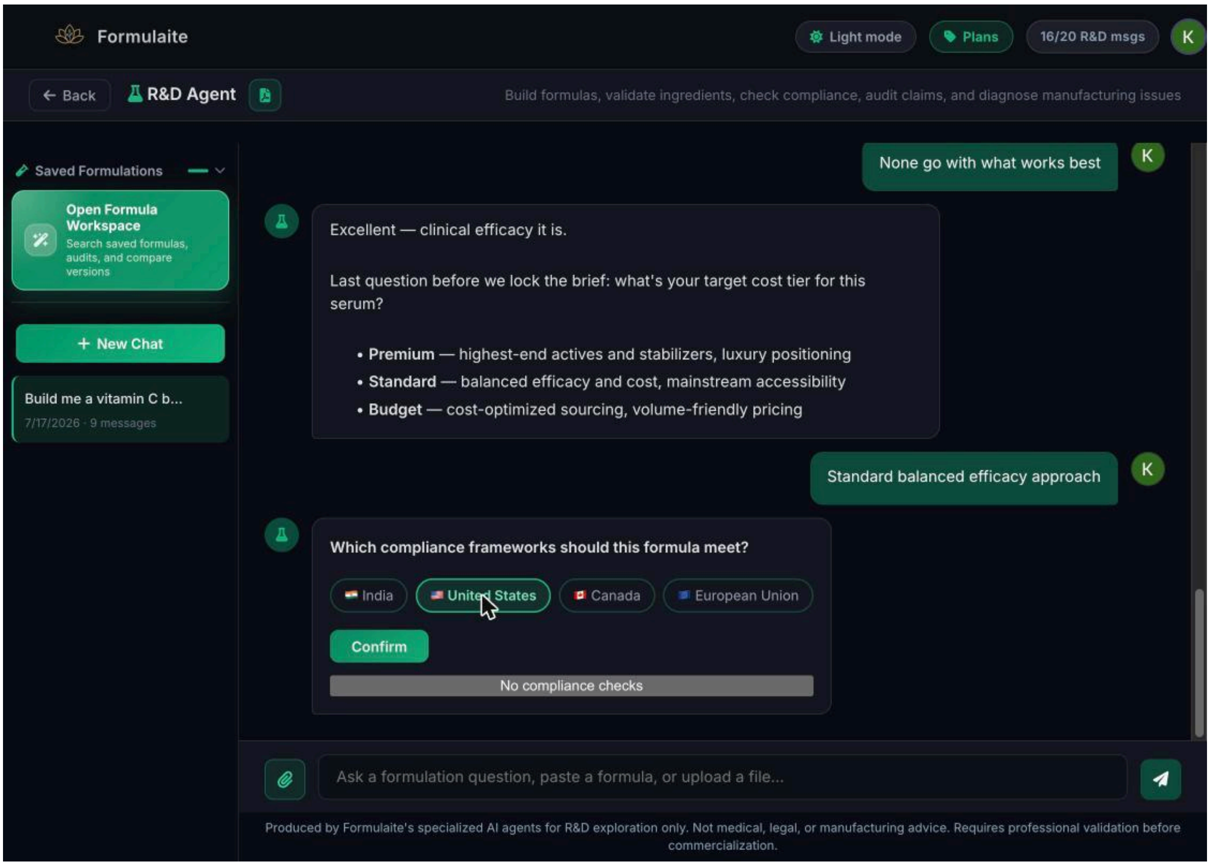

Figure 3. The compliance-framework question uses multi-select flag toggles plus a Confirm button, instead of the usual quick-reply chips — the widget type adapts to the data being asked for.

4.3 Transparent, staged asynchronous generation

Once the brief is locked in, the tool does not show a generic spinner. It transitions into a visibly staged “reasoning” log with sequential, labeled steps: Literature review, Paper screening, Evidence mapping, Actives selection, Dose tuning, and Compliance checking, each with a live one-line status description (e.g. “Reviewing journal literature on Rumex occidentalis clinical trial humans…”). Earlier steps collapse into a “Show earlier steps” accordion so the log never overwhelms the chat. A persistent banner reads “This may take a few minutes” with a “Notify me when ready” option, letting the user step away rather than stare at a loader. In this test run, the full pipeline took roughly two minutes end-to-end.
This is arguably the strongest UX element of the tool: it turns what could feel like an unresponsive black box into a legible, trustworthy pipeline, and it is honest about being slow rather than faking instant results.

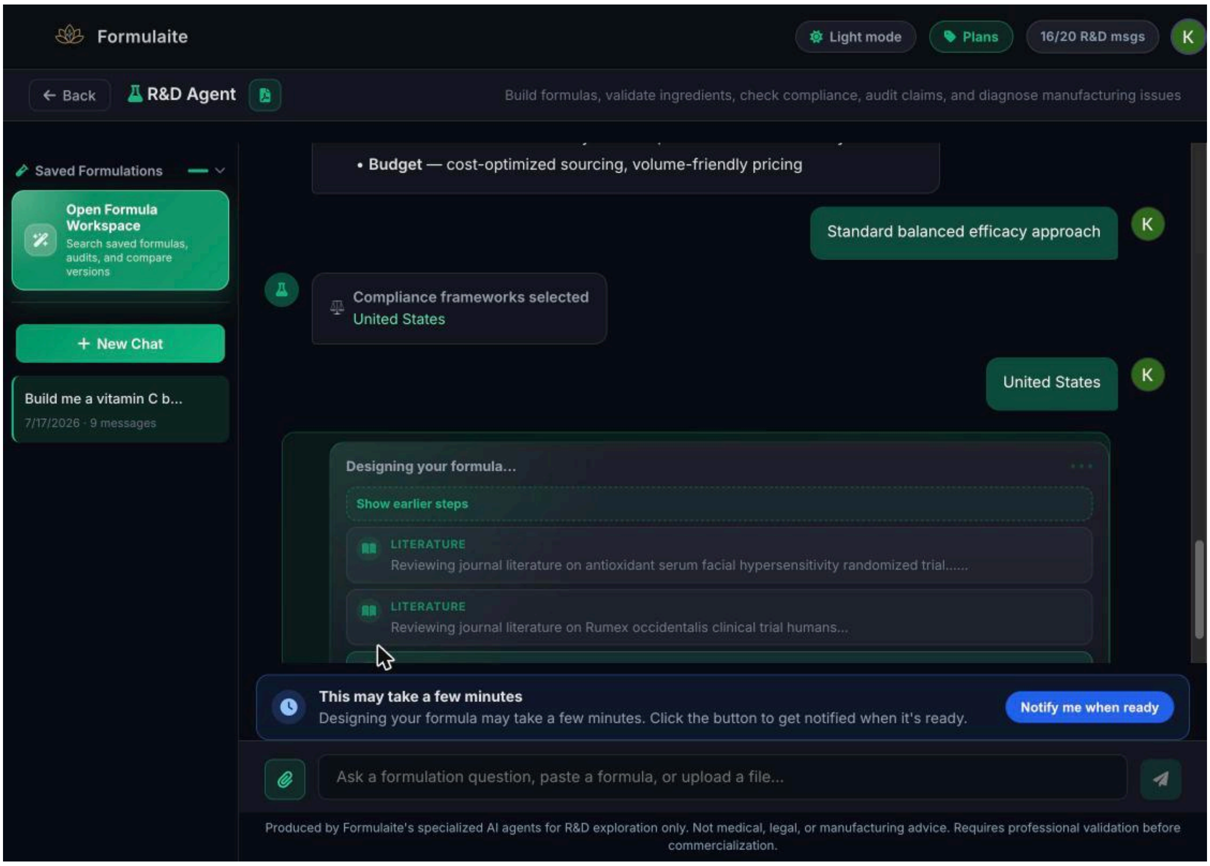

Figure 4. Early pipeline stage: live “Literature” status lines plus a “This may take a few minutes / Notify me when ready” banner instead of a generic spinner

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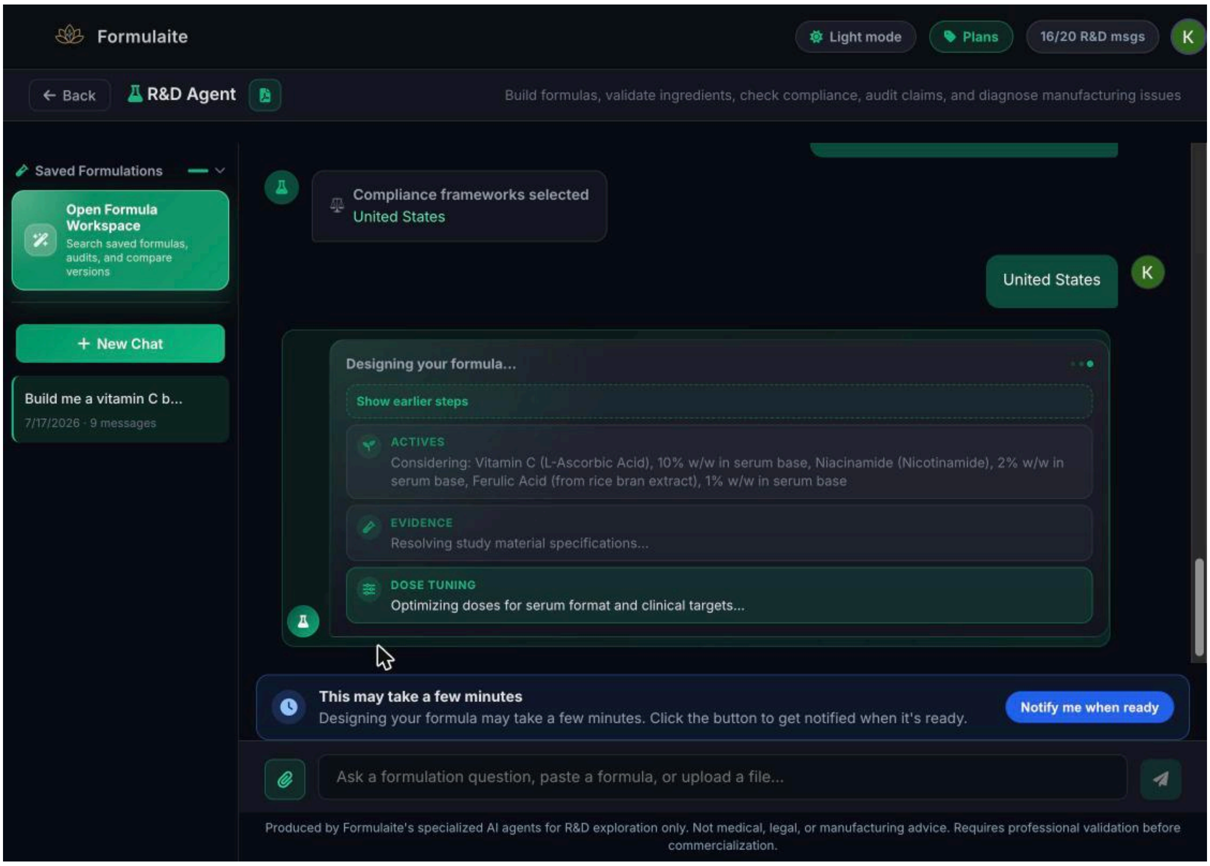

Figure 5. A later pipeline stage: candidate actives now shown with real concentrations (e.g. “Vitamin C (L-Ascorbic Acid), 10% w/w”), moving into dose tuning.

4.4 The final formula output

The finished result is delivered as a structured card, not a plain paragraph: a header (product type, ingredient count, and a green “Compliant” badge), followed by a per-ingredient list showing name, exact concentration (e.g. “Vitamin C (L-Ascorbic Acid) 10% w/w in serum base”), a per-application dose (“50 mg per 0.5 mL serum application”), and a small “DOI” citation link. Each row is collapsed to one line by default and expands via a chevron to reveal the full clinical rationale behind that concentration. Below the ingredient table, the agent also writes a short marketing-style description of the formula and offers targeted, one-tap iteration chips (e.g. “Increase Vitamin C to 15% for deeper brightening,” “Add hyaluronic acid for hydration boost,” “Switch to stabilized Vitamin C derivative”). This is the actual “blending” mechanic: adjustments happen by asking the agent to regenerate with a new constraint, not by dragging a percentage slider or typing into a spreadsheet-style grid.

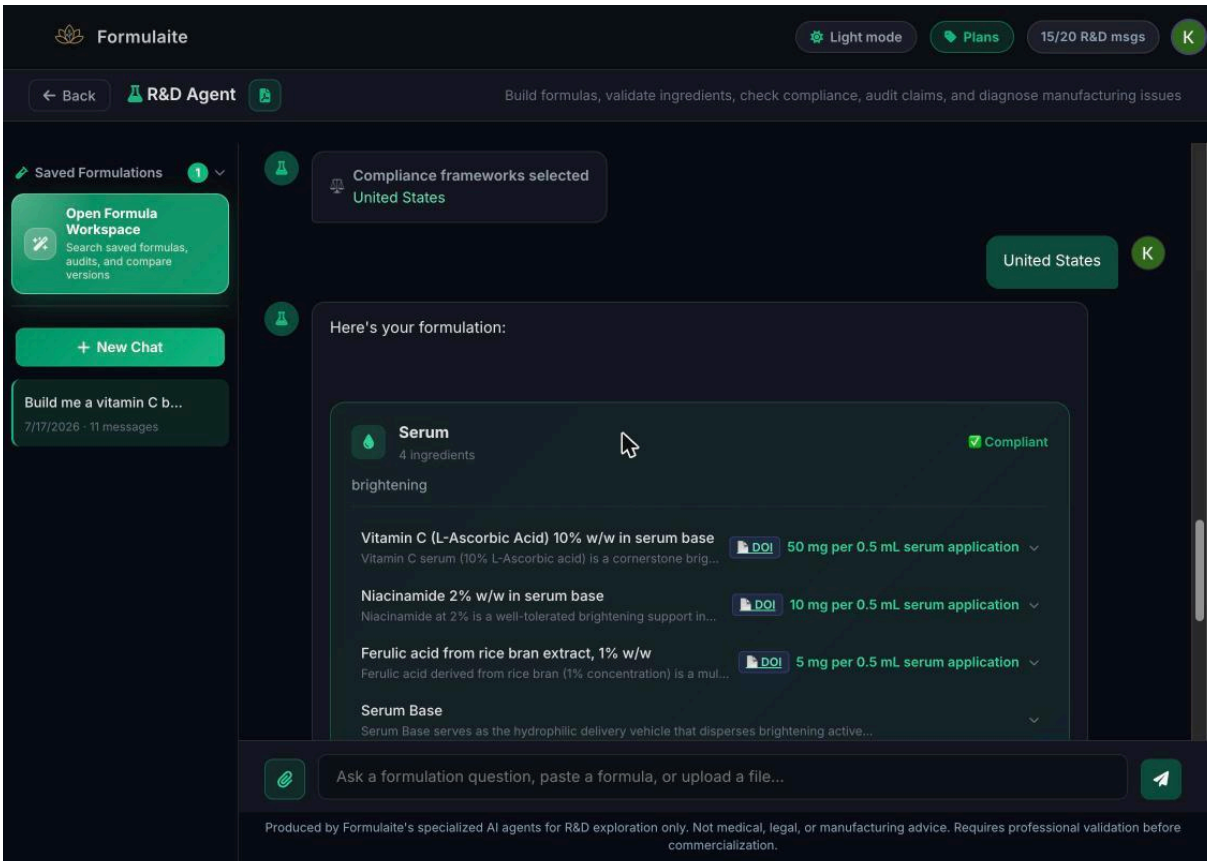

Figure 6. The finished formula card: header with a "Compliant" badge, and an ingredient table with concentration, per-application dose, and a DOI citation link for each ingredient.

4.5 Evidence-linked ingredient detail

Clicking the chevron on any ingredient row expands it into a full clinical rationale rather than generic marketing copy. Expanding the Vitamin C row surfaced a specific citation: a 12-week randomized controlled trial in 66 healthy Chinese females using a 10% vitamin C serum, with quantified outcomes (e.g. reductions in advanced glycation end-products, carbonylation, and skin yellowing/redness, all statistically significant). The "DOI" label next to each ingredient links out to the primary source — in this case, a real Wiley Online Library article, Journal of Cosmetic Dermatology, "Study on the Multiple Efficacies of Vitamin C Serum in Anti-Glycation, Anti-Carbonylation, Antioxidation, and Anti-Inflammation of Human Skin Based on In Vivo Tests" (onlinelibrary.wiley.com/doi/10.1111/jocd.70888). Linking directly to a peer-reviewed, DOI-indexed journal article (rather than a blog post or an unverifiable in-house claim) is a meaningful credibility signal for a scientific/regulated audience, and it lets a formulator actually check the underlying evidence before trusting a concentration.

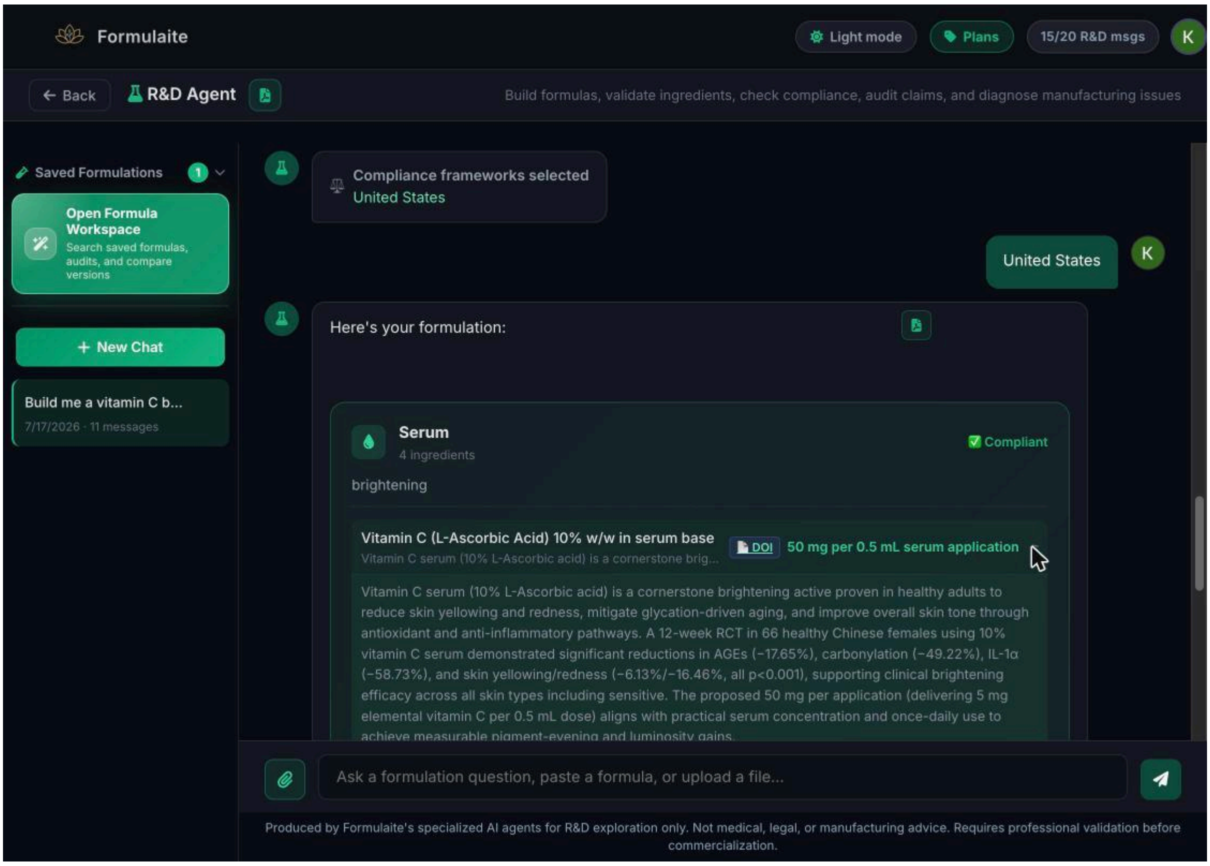

Figure 7. An expanded ingredient row for Vitamin C, showing the specific RCT (66 subjects, 12 weeks) cited to justify the 10% w/w concentration, with a DOI link to the source article.

4.6 Iteration and the Saved Formulations workspace

Saved work lives in a "Saved Formulations" panel in the chat sidebar, which opens into a full-screen "Formula Workspace" modal with a search bar ("Search by benefit, ingredient, audit summary, or project"), tab filters (All / Formulations / Audits), and a compare feature that lets a user select 2-4 saved formulations to diff side by side. This is a solid, if fairly standard, project-management layer bolted onto the chat, and it is the closest thing the product has to a persistent "library" view outside the linear conversation.

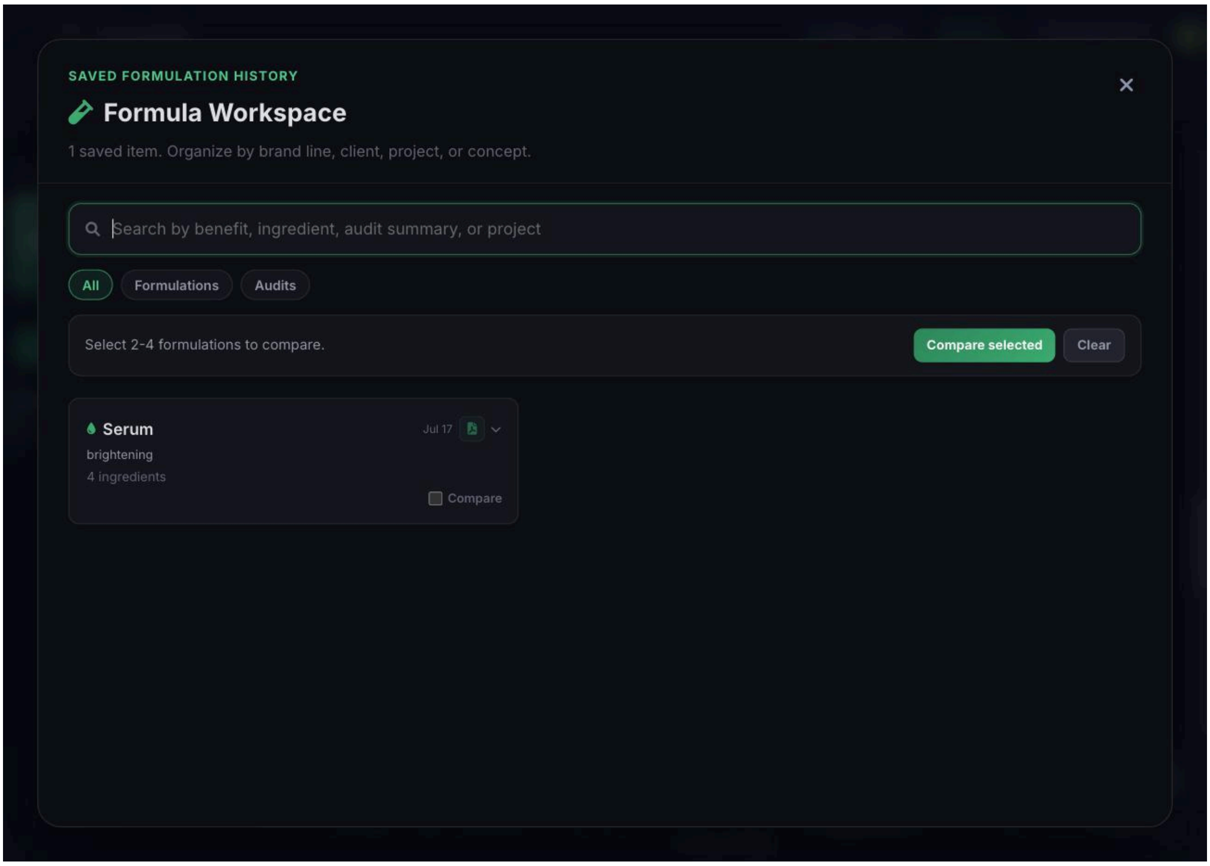

Figure 8. The "Formula Workspace" modal: search, All/Formulations/Audits filters, and a 2-4 item comparison picker.

5. Evidence Sourcing & Citations

As covered in Section 4.5, every generated ingredient links to a "DOI" tag. In the run tested, this resolved to a genuine, currently indexed Wiley Online Library article in the Journal of Cosmetic Dermatology (onlinelibrary.wiley.com/doi/10.1111/jocd.70888), not a placeholder or a dead link. From a UX standpoint this is a deliberate, high-value design decision: it moves the tool from "AI says this concentration is good" to "here is the peer-reviewed study this concentration is drawn from," which is exactly the kind of traceability a regulated-industry formulator needs before taking a number into a lab or a filing. It would be even stronger if the citation preview appeared inline (e.g. authors, journal, year) rather than requiring a click-through, since right now the only way to sanity-check a source is to leave the app entirely.

6. The "Connect to Manufacturer" Upsell & Its Email Behavior

A recurring promotional banner, "Matched low-MOQ GMP partner for your Serum / Formulaite connects you for free — vetted partner, no broker fees," appears at least twice in a single session: once immediately after the first message is sent (before any formula exists), and again attached to the finished formula card. It is styled in the same reassuring

green as genuine system feedback (see Section 2), which blurs the line between the tool's scientific output and a lead-generation unit sitting inside the same scroll stream.
Clicking the banner's "Connect to Manufacturer for free" link does not open an in-app request form or contact a third-party manufacturer directly. It opens a pre-filled Gmail compose window via a mailto link, addressed to team@formulaite.ai (Formulaite's own team, not an external manufacturer), with the subject "GMP manufacturing intro — Serum formulation" and a body that Formulaite auto-generates from the session's data: delivery format, product type, ingredient count, the benefit/goal, the full formula description and ingredient list generated earlier in the chat, an MOQ preference line, and a "Formulaite run ID" plus a hidden routing footer (route=default; cta=connect; profile=gmp_profile_default). In other words, the "free manufacturer connection" is actually a lead captured by Formulaite itself, which presumably forwards qualifying requests to its own vetted partner network afterward. This is not disclosed anywhere in the banner copy, and a user could reasonably believe they are emailing a manufacturer directly rather than submitting a sales lead to Formulaite.
For reference, the generated email observed during this test read as follows:
To: team@formulaite.ai | Subject: GMP manufacturing intro— Serum formulation

Hidden footer: Formulaite routing: route=default; cta=connect; profile=gmp_profile_default